

Supporting Information

Spatial structure before algorithmic complexity: evidence based guidance for marine species distribution modelling where data are limited

S1 Search strategy and screening

S1.1 Databases and coverage

Searches combined method terms (Random Forest, boosted regression trees, VAST, sdmTMB, generalised additive model, MaxEnt, joint species distribution model, graph neural network), spatial terms (spatial autocorrelation, spatial cross validation, geostatistical, random field), data limitation terms (data limited, small sample, sparse) and marine ecosystem descriptors (marine, ocean, estuarine, fish, cetacean, benthic). Complete Boolean strings are in Table S2.

Table S1: Records identified, screened and included by source. Counts are unique records after duplicate removal within each source at each stage. Thirteen studies eligible at full text were duplicates across sources and were resolved at the final stage, so the per source included counts sum to 108 before that resolution and 95 after it.

Source	Period	Identified	After title/abstract	After full text	Included
Web of Science	2001–2025	185	89	58	50
Google Scholar	2001–2025	95	48	28	25
Discipline specific journals	2010–2025	35	25	14	13
Reference tracking	—	15	10	8	8
Total		330	172	108	95

Table S2: Complete Boolean search strings by topic area. Strings were executed across Web of Science and Google Scholar; the discipline specific journal search used the same topic terms applied to journal level search interfaces.

#	Topic area	Boolean search string	Rationale
1	General Spatial ML/SDM	("spatial machine learning" OR "species distribution model*" OR "habitat model*" OR "niche model*" OR "SDM" OR "HSM") AND (marine OR ocean* OR aquatic OR fish* OR cetacean* OR "marine mammal*" OR benthos OR benthic OR pelagic OR demersal OR "marine ecosystem*")	Broad capture of spatial modeling approaches in marine systems
2	Random Forest	("random forest*" OR "RF" OR "ensemble learning" OR "random tree*") AND (marine OR ocean* OR fish* OR cetacean* OR aquatic) AND (spatial OR distribution OR habitat OR predict*)	Specific RF method with marine applications
3	Boosted Regression Trees	("boosted regression tree*" OR "BRT" OR "gradient boosting" OR "GBM" OR "boosted tree*" OR "gradient boost*") AND (marine OR ocean* OR fish* OR "marine species" OR cetacean*) AND (distribution OR habitat OR spatial)	BRT and gradient boosting variants
4	VAST Framework	("VAST" OR "vector autoregressive spatio-temporal" OR "vector autoregressive spatiotemporal" OR "geostatistical delta" OR "spatio-temporal GLMM") AND (fisheries OR marine OR ocean* OR fish OR "stock assessment")	Purpose-built marine spatio-temporal framework
5	Spatial Cross-Validation	("spatial cross-validation" OR "spatial cross validation" OR "spatial CV" OR "spatial autocorrelation" OR "spatial validation" OR "spatial blocking") AND ("species distribution" OR "habitat model*" OR SDM OR "predictive model*" OR "machine learning")	Critical validation methodology
6	GAM Spatial Models	("generalised additive model*" OR "GAM" OR "additive model*") AND spatial AND (marine OR ocean* OR fish* OR aquatic) AND (distribution OR habitat)	Traditional/hybrid spatial modeling approach
7	MaxEnt	("MaxEnt" OR "maximum entropy" OR "Maxent") AND (marine OR ocean* OR aquatic OR fish*) AND (species OR distribution OR habitat OR SDM)	Presence-only modeling dominant method
8	Transfer Learning	("transfer learning" OR "extrapolation" OR "transferability" OR "model transfer*" OR "spatial transfer*" OR "temporal transfer*") AND (species OR marine OR ecology OR distribution OR SDM) AND (predict* OR model*)	Transferability and forecast skill assessment
9	Data-Limited Contexts	("data limited" OR "data-limited" OR "data poor" OR "limited sample*" OR "small sample*" OR "sparse data") AND (marine OR fisheries OR ocean*) AND (model* OR predict* OR distribution OR SDM)	Explicit focus on data constraints
10	Graph Neural Networks	("graph neural network*" OR "GNN" OR "deep learning" OR "neural network*" OR "convolutional neural") AND (marine OR ocean* OR species OR ecology OR spatial) AND (distribution OR habitat OR predict*)	Emerging deep learning methods
11	Uncertainty Quantification	("uncertainty quantification" OR "prediction uncertainty" OR "model uncertainty" OR "adaptive monitoring" OR "value of information") AND ("species distribution" OR marine OR ecology OR SDM) AND (spatial OR predict*)	Uncertainty and adaptive approaches

#	Topic area	Boolean search string	Rationale
12	Spatial Random Forest Specifically	("spatial random forest" OR "spatial RF" OR "RFsp" OR "geographically weighted random forest") AND (ecology OR species OR distribution OR marine OR spatial)	Enhanced spatial RF implementations
13	Ensemble Methods	(ensemble OR "model averaging" OR "multi-model" OR "stacked model*") AND ("species distribution" OR SDM OR marine OR ecology) AND (predict* OR spatial)	Multiple model combinations
14	Validation Reviews	("cross-validation" OR "model validation" OR "performance assessment") AND spatial AND (autocorrelation OR blocking OR structure) AND (ecology OR species OR SDM)	Methodological validation literature
15	Joint Species Distribution Models	("joint species distribution" OR "JSDM" OR "multivariate species" OR "HMSC" OR "hierarchical modelling of species communities" OR "community-level model") AND (marine OR ocean* OR fish* OR aquatic) AND (spatial OR distribution OR predict*)	Joint modeling approaches leveraging on data sharing mechanisms

S1.2 Inclusion and exclusion criteria

Included: marine or estuarine application; a spatial modelling component; sample size reported; performance metrics or documented validation practice; peer reviewed and in English.

Excluded: purely terrestrial systems, except where a study provided a methodological result with no marine equivalent, in which case it was retained and flagged; reviews without original analysis; insufficient methodological detail to determine the spatial treatment or validation design; no performance assessment of any kind.

Corpus membership and citation are separate. The corpus comprises studies that passed the criteria above and contributed evidence to the synthesis. The reference list additionally contains statistical, algorithmic and software sources that have no study system at all, and which are cited for provenance or implementation rather than as evidence: these were never screened as corpus candidates and are not counted in any total reported in this review. Table S3 separates the two, and distinguishes within the retained non-marine set between studies that anchor a deliverable and studies cited only for method provenance.

Table S3: Studies applied outside marine and estuarine systems, or to simulated data, retained under the methodological exception in Section S1.2. Each is identified as non-marine at every point of use in the main text, and the consequence for the evidence base is addressed in Section 4.3. The classification basis records how the system was determined: *verified* against the publisher record, from the article *title*, or from the *journal* scope.

Study	System	Reason for retention	Findings supported	Basis
<i>Studies anchoring a deliverable</i>				
Hengl et al. (2018)	Soil and environmental benchmarks, terrestrial	The only study in the review that crosses spatial treatment against algorithm family on identical data. Its kriging comparison also prevents D1 being read as an endorsement of tree ensembles	D1 (Section 3.2); Table 4 of the main text	Title
Talebi et al. (2022)	Geoscience	Independent corroboration of the same contrast in a second domain, reported on error rather than discrimination metrics	D1 (Section 3.2)	Journal
Mushagalusa et al. (2024)	Simulated abundance data; no study system	Compares standard, design based, buffered, environmentally blocked and spatially blocked partitioning on data generated from a Gaussian process. Simulation is the only setting in which the true surface is known, so validation optimism can be measured rather than inferred	D1, D3 (Sections 3.2, 3.4)	Verified
Ploton et al. (2020)	Tropical forest biomass, terrestrial	The reference demonstration that random partitioning inflates apparent accuracy under spatial autocorrelation. Its 11.8 million observations have no marine counterpart	D3 (Section 3.4)	Title
Mi et al. (2017)	Cranes, wetland	One of only two studies quantifying performance below 100 observations. No marine study in the corpus reports performance at comparable sample sizes	D2 (Section 3.3); Tables 3 and 4 of the main text	Title
<i>Studies cited for method provenance or implementation</i>				

Study	System	Reason for retention	Findings supported	Basis
Behrens et al. (2018)	Soil, terrestrial	Introduces Euclidean distance fields, the spatial feature construction on which the tree ensemble rows of Table 1 of the main text depend	Table 1 of the main text	Journal
Fouedjio and Arya (2024)	Geoscience	Locally varying geostatistical machine learning; one of the options for relaxing stationarity	Section 3.1	Journal
Elith et al. (2006)	226 species across six regions, terrestrial	The benchmark comparison for presence-only methods. Cited for provenance; its data rich setting is the regime this review distinguishes itself from	Table 1 of the main text	Title
Prasad et al. (2006)	Tree species, terrestrial forest	Early comparison of bagging and boosting for ecological prediction; provenance only	Section 3.1	Journal
<i>Spatial statistics and machine learning sources with no study system</i>				
Saha et al. (2023)	Statistical theory	The formal treatment of Random Forest under spatial dependence, cited as the dependent data formulation	Table 1 of the main text	Journal
Du et al. (2020)	Spatial statistics	Geographically neural network weighted regression; non-stationarity options	Section 3.1	Title
Yang et al. (2023)	Spatial statistics	Geographically weighted regression combined with machine learning	Section 3.1	Journal

S1.3 Deliverable coding

Two distinct quantities describe how the corpus supports the four deliverables. The first is the number of distinct sources the main text cites in support of each, which is fixed by the manuscript and reported in Table S4. The second is the number of corpus studies coded as contributing evidence to each, which is recorded alongside study attributes in the extraction records and is the larger of the two, since a study may inform a finding without being cited where that finding is stated.

Table S4: Distinct sources cited in support of each deliverable, counted from the citation commands in the main text and attributed to the section in which they appear. A source cited under more than one deliverable is counted under each, so the column does not sum to the 57 distinct sources cited in the main text, and neither figure is the corpus total. These counts are a property of the argument as written and can be checked against the printed paper; they are a lower bound on corpus contribution, since a study may inform a finding without being cited at the point the finding is stated. Marine status follows the classification in Table S3.

Deliverable	Sources cited	Of which marine	Of which crossing both contrasts
D1 spatial structure before algorithmic complexity	6	3	3
D2 coverage before sample size	6	4	0
D3 validation matched to inference target	4	3	0
D4 decision framework and applications	15	15	0
Method families (Table 1 of the main text)	21	17	1
Quantitative claims (Table 4 of the main text)	9	7	1

The pattern in Table S4 is itself informative and is discussed in Section 4.3 of the main text. Support for D1 rests on six sources, only three of them marine, and all three studies crossing spatial treatment against algorithm family on the same data are non-marine. D2 and D3 rest on small numbers of sources reporting the relevant experiment directly, which is a statement about the state of the literature rather than about the strength of the direction they agree on. D4, by contrast, draws on fifteen sources, all marine, because it synthesises applied work rather than resting on comparative experiments.

S2 Extraction protocol, extracted studies and quality assessment

S2.1 Selection for quantitative extraction

Twelve studies were selected on the basis of at least one of four criteria: a direct method comparison on identical data; an explicit assessment of validation optimism; a sample size experiment with associated performance metrics; or an authoritative performance benchmark for a specific method. Selection was made before extraction and was not conditioned on the direction of the reported result.

S2.2 Extraction fields

Sample characteristics. Number of observations, spatial extent, temporal extent, sampling design, response type, prevalence where applicable.

Methodological detail. Algorithms compared, software and version, predictor set, spatial treatment, model complexity settings.

Validation. Partitioning scheme, block size or temporal unit where applicable, whether an independent test set was available, whether the inference target was stated.

Results. Metrics reported, values, direction of the contrast, whether the comparison was within study.

Context. Taxon, ecosystem type, management application if any.

Table S5: Studies selected for detailed quantitative extraction.

Study	<i>n</i>	Taxon	System	Comparison	Metrics reported	Result as reported	Quality
Becker et al. (2020)	4,713	Cetaceans	Pelagic, California Current	BRT against GAM; <i>k</i> fold against leave one year out	Area under the ROC curve; rank correlation	0.85 against 0.83; density surface GAM better on rank correlation; 0.11 partitioning difference	High
Luan et al. (2020)	40–120	Fish	Demersal	Random Forest across sample sizes and sampling designs	Performance stability	Distributed samples outperform clustered samples of equal or greater size	High
Mi et al. (2017)	33–75	Cranes	Wetland	Random Forest against MaxEnt and BRT	Area under the ROC curve	0.82–0.89	High
Thorson et al. (2015)	28 species	Groundfish	Demersal	Geostatistical delta GLMM against design based estimation	Standard error of index	Approximately 20% reduction	Very high
Thorson (2019a)	~300 per year	Groundfish	Demersal	Retrospective forecast skill by horizon	Correlation	$r = 0.6$ – 0.7 at one year; 0.3 – 0.4 at five years	High
Hengl et al. (2018)	multiple benchmarks	Soil	Terrestrial	Spatial Random Forest against kriging	R^2	Approximately 0.41 against approximately 0.60 on Meuse	High
Ploton et al. (2020)	11.8 million	Trees	Terrestrial	Random against spatially structured validation	Conceptual and empirical	Ignoring autocorrelation inflates apparent accuracy	Very high
Mushagalusa et al. (2024)	simulated	Simulated	Terrestrial	Random against spatial cross validation	Error magnitude	Spatial partitioning yields larger, more honest errors	High
Grüss et al. (2021)	~300 per year	Walleye pollock	Demersal, eastern Bering Sea	Spatio-temporal model with environmental covariates	Variance explained; correlation	Cold pool approximately 40%; cross validation $r = 0.72$	High
Zelli et al. (2025)	1,247	Corals and sponges	Deep sea, New Zealand	BRT with climate projection	Projected habitat retention	Approximately 23% retained under high emissions	High

Study	<i>n</i>	Taxon	System	Comparison	Metrics reported	Result as reported	Quality
Roberts et al. (2017)	review	Multiple	Multiple	Cross validation framework	Methodological	Basis for partitioning matched to inference target	Very high
Flynn and Marshall (2013)	multiple	Lanternfish	Pelagic	GAM with MaxEnt	Application	Deep pelagic application under sparse sampling	Moderate

S2.3 Quality assessment

Studies were rated across five dimensions: sample adequacy for the inference attempted; validation rigour; reporting clarity, including whether the inference target was stated; statistical robustness; and relevance to data limited marine settings. Very high denotes exemplary performance on all five; high denotes four or five met with minor limitations; moderate denotes three met adequately. Studies rated low were excluded from quantitative extraction and retained, where relevant, for narrative synthesis only.

Quality ratings determined whether a study could anchor an attributed numerical claim in the main text. They were not used as weights, since no pooling or weighted averaging was performed at any stage.

S3 Performance metrics and the comparison protocol

S3.1 Metrics encountered in the corpus

Discrimination metrics. The area under the receiver operating characteristic curve measures the probability that a randomly chosen presence receives a higher predicted score than a randomly chosen absence, bounded between 0.5 and 1. Its dominance reflects three convenient properties: independence from any threshold choice, boundedness across study contexts, and near universal implementation. Three limitations bear on synthesis. It is insensitive to calibration, so a model can discriminate well while producing biased probabilities. It depends on the spatial extent from which absences or background points are drawn, so values are not comparable across studies with different domains. It weights omission and commission errors equally, which rarely matches a conservation decision context (Lobo et al., 2008; Guillera-Arroita et al., 2015). The true skill statistic partially addresses prevalence dependence (Allouche et al., 2006).

Error metrics. Root mean square error and mean absolute error are scale dependent and not comparable across systems without normalisation. R^2 is bounded but depends on the variance structure of the data and on whether it was computed in training or in validation.

Management relevant metrics. For abundance and density applications, agreement between a model derived index and an independent estimator, expressed as a correlation or a relative standard error, is often more decision relevant than any pointwise predictive metric.

S3.2 Why no pooled estimate is reported

No defined transformation relates a difference in a discrimination metric to a difference in an error metric. Any mapping between them requires assumptions about prevalence, the shape of the predicted score distribution and the loss function, none of which is recoverable from a published summary. Averaging across studies that report different metric families therefore produces a quantity with no interpretation, and averaging within a metric family across studies with different taxa, prevalences, domains and validation designs produces a quantity whose reference population is undefined.

An earlier version of this synthesis reported cross study ranges for the spatial structure effect and the algorithm choice effect, expressed in the area under the curve. Those ranges were assembled by treating values reported in different metric families as approximately commensurate and averaging across them with weights reflecting study size and quality. That procedure has no defensible basis, and the resulting values have been withdrawn rather than reframed. Appendix S4 records each withdrawn claim, the sources previously cited in its support, and what those sources actually report.

S3.3 The protocol applied

1. Within study contrasts are reported with attribution to the source, its taxon, its system and the metric that source used.
2. No value is averaged, weighted or otherwise combined across studies.
3. Where studies agree in the direction of a contrast but not in metric or magnitude, the direction is reported together with the number of studies supporting it, and no magnitude is attached.
4. Every numerical value in the main text appears in Table 4 of the main text with its source and its scope of generalisation.
5. Ranges, where they appear, describe the values reported by named studies and are not confidence intervals.

This protocol is a response to the current state of the evidence base. It is not offered as a general method, and it does not solve the underlying problem, which requires standardised multi-metric reporting in primary studies.

S4 Claim traceability audit

Every quantitative claim considered during preparation was traced to its primary source and checked against what that source reports. Claims that could not be traced were removed.

Table S6: Method families applied to marine species distribution modelling under data limitation. Indicative data requirements are lower bounds reported under favourable conditions and should be read alongside the coverage requirement in Section 3.3.

Family	Spatial treatment	Response types	Indicative minimum data	Uncertainty	Cost	Principal limitations	Key sources
Random Forest, spatial variants	Buffer distances, Euclidean distance fields, spatial lags supplied as predictors	Presence absence, abundance	~50 to 100 observations for classification; more for abundance	Ensemble spread only; no spatial variance component	Low; scales well	Weak extrapolation; rotation dependence of coordinate encoding	Breiman (2001); Hengl et al. (2018); Behrens et al. (2018); Sekulić et al. (2020); Saha et al. (2023)
Boosted Regression Trees	As above; no native spatial term	Presence absence, abundance	~100 lower bound; ~200 for interactions	None native	Low to moderate	Tuning sensitive; overfits at small n without careful regularisation	Friedman (2001); Elith et al. (2008); Becker et al. (2020)
Geostatistical GMRF models (VAST, sdmTMB, tinyVAST)	Latent Gaussian random fields on an SPDE mesh; spatial and spatio-temporal	Presence absence, abundance, biomass, composition; delta models for zero inflation	~30 stations across ≥ 3 years for basic configurations; 100 to 500 mesh nodes	Full posterior or Laplace approximated; propagated to derived quantities	Moderate to high	Mesh specification requires judgement; scales super-linearly with mesh complexity	Thorson et al. (2015); Thorson (2019b); Anderson et al. (2025); Thorson et al. (2025); Kristensen et al. (2016)
Spatial GAMs	Tensor product and thin plate smooths of coordinates	Presence absence, abundance, density surfaces	Comparable to BRT; stable at small n with penalisation	Classical intervals; smoothing parameter uncertainty available	Low	Functional form partly pre-specified; smooth basis choice affects extrapolation	Wood (2017); Miller et al. (2022); Becker et al. (2020)
MaxEnt	Background sampling defines the spatial domain; no explicit dependence term	Presence only	Workable at small n ; sensitive to background definition	Limited	Low	Performance depends strongly on background extent; presence only inference constraints	Phillips et al. (2006); Elith et al. (2011)

Family	Spatial treatment	Response types	Indicative minimum data	Uncertainty	Cost	Principal limitations	Key sources
Joint species distribution models	Species correlation structure; spatial latent variables in some implementations	Community presence absence or abundance	Adequate replication per species; benefit concentrated with ≥ 10 co-occurring species	Full posterior in Bayesian implementations	High; MCMC in hours	Community sample size requirements; limited benefit when species share few associations	Warton et al. (2015); Ovaskainen et al. (2017); Norberg et al. (2019)

S5 Limitations of the review

Table S7: Limitations and mitigation.

Limitation	Consequence	Mitigation applied	Residual risk
Metric heterogeneity across studies	Formal meta-analysis not possible	Directional synthesis with attributed within study values only	Findings are directional; magnitudes unquantified
Primary support for the spatial structure finding is largely non marine	Marine magnitude unknown	Terrestrial and simulation studies flagged at every point of use; marine corroboration reported separately	Transfer of the mechanism is expected but unconfirmed for marine systems
Publication bias toward positive comparative results	Frequency of method failure not estimable	No claim is made about how often any approach fails	Corpus overstates the consistency of comparative advantage
English language, peer reviewed sources only	Grey literature and non English work excluded	Stated	Regional practice under-represented
Inference target rarely stated in primary studies	Reported performance figures often uninterpretable	Studies coded for whether the target was stated; count reported in Section 3.4	Some reported values may reflect a different target than assumed
Rapidly moving software landscape	Implementation guidance dates quickly	Software table gives versions and access dates	Appendix S7 will require updating

S6 Comparison with previous reviews

Table S8: Contribution relative to previous reviews of marine species distribution modelling and machine learning in marine systems.

Review	Scope	Quantitative synthesis	Data limitation addressed	Validation guidance	Minimum data thresholds
Robinson et al. (2017)	Marine SDM practice	No	No	Partial	No
Melo-Merino et al. (2020)	ML in marine systems	Bibliometric	No	No	No
Rubbens et al. (2023)	ML in marine ecology	No	No	Partial	No
Pasanisi et al. (2024)	Marine SDM	No	No	No	No
Klaassen et al. (2025)	Methodological trends	No	No	No	No
This review	Marine SDM under data limitation	Directional, with attributed within study values	Yes, as the organising scope	Yes, matched to inference target	Yes, method specific

S7 Context specific guidance by environment and objective

Table 3 of the main text gives minimum data requirements by method. Those thresholds are not invariant across marine environments or across the use to which a model will be put, and this appendix gives the adjustment.

The scaling factors in Table S9 are author estimates informed by patterns in the reviewed literature. They are not empirically derived multipliers from controlled experiments, no study in the corpus reports them, and they should be read as an ordering of environments by expected data requirement rather than as coefficients. They are recorded as declared estimates in Appendix S4 so that the audit covers every number in this review.

Table S9: Environment specific scaling of minimum sample size requirements. Multipliers are author estimates, not empirical coefficients.

Environment	Scaling factor	Random Forest minimum	Geostatistical minimum	Rationale
Pelagic	$\sim 1.5\times$	~ 75 observations	~ 45 stations	High spatio-temporal variability from currents, mesoscale eddies and frontal dynamics; mobile species with large ranges; weaker habitat associations
Demersal and benthic	$1.0\times$ (reference)	~ 50 observations	~ 30 stations	Stronger habitat structure and more predictable species–substrate associations
Coastal	$\sim 0.75\times$	~ 40 observations	~ 25 stations	Accessible sampling, strong environmental gradients, greater data availability
Deep sea	$0.6\text{--}1.0\times$	30–50 observations	Not recommended	Extreme data limitation; preference absence over abundance, and methods that function at minimal sample sizes

The ordering rests on the variability of the environment rather than on accessibility alone. Pelagic systems weaken habitat associations and inflate residual variability (Bell et al., 2014; Robinson et al., 2011); demersal and benthic systems supply the reference case; coastal work benefits from accessibility and steep gradients (Canonico et al., 2019). Deep-sea applications face the severest limitation, and the appropriate response is to narrow the inferential target, predicting occurrence rather than abundance and coarse pattern rather than fine surfaces, rather than to relax the performance standard a model is held to.

S7.1 Index standardisation and two stage assessment workflows

One class of application warrants separate treatment, because the appropriate method choice is determined by a downstream consumer rather than by the prediction task itself. In two stage assessment workflows a model is fitted to survey data to produce a time series of annual abundance indices, and those indices are then passed to a separate assessment model that combines them with catch data, biological information and population dynamics assumptions (Thorson, 2019b). The structural assumptions of the downstream consumer constrain the upstream choice.

The binding constraint is the temporal error structure of the indices. Assessment models generally treat the index as a series of observations with independent year specific errors, and their likelihood, goodness of fit diagnostics, retrospective analysis and uncertainty propagation all rest on that independence. A model that imposes strong temporal autoregressive structure on the abundance field produces indices that are correlated across years, which are not plausible draws from independent year specific errors. The assessment then understates true uncertainty, because correlated index errors appear more informative than they are, and

correlation propagates into estimated trends as retrospective pattern.

This is the reason for a specific structural choice in the geostatistical packages in Table 1 of the main text (Thorson, 2019b). They permit rich spatial structure through Gaussian Markov random fields on the spatial random effects, which is what makes them useful, while not imposing strong temporal autoregression on the year to year evolution of the abundance field. The annual indices that result are spatially informed and temporally independent in their error structure. Methods that impose explicit AR(1) or random walk structure on the abundance field yield smoother indices that look more attractive in isolation and are less appropriate for this workflow.

Two qualifications. The recommendation applies to two stage workflows only. Where the model is itself the final product, in habitat mapping, protected area design, dynamic management or short term forecasting, temporal autoregressive structure may improve prediction at the cost of an independence no downstream consumer requires. And the recommendation concerns temporal structure, not spatial: spatial structure should be included throughout, per D1. Practitioners should establish whether model output will feed an assessment before committing to a method's structural choices.

Table S10: Objective–method pairings. Minimum data entries carry the same status as their sources: the geostatistical requirement is reported by Thorson (2019b) and the joint model requirements follow Norberg et al. (2019); the remainder are author guidance consistent with Table 3 of the main text rather than values reported by any single study.

Objective	Recommended methods	Indicative minimum data
Protected area habitat identification	Geostatistical model with uncertainty maps, or spatial Random Forest	~100 observations covering the candidate extent
Abundance index for direct interpretation	Geostatistical spatio-temporal model with a multi-year series	~30 stations across ≥ 5 years
Abundance index feeding a two stage assessment	Geostatistical models in default configuration, temporal shrinkage off	30 stations across ≥ 3 years (Thorson, 2019b)
Climate projection	Constrained ensemble; mechanistic and correlative combined	~300 observations spanning the environmental range
Biodiversity hotspot mapping	Multi-species geostatistical model, stacked single species models, or a joint model with spatial latent factors	~50–100 per species, or 20–30 per species across ≥ 50 sites with ≥ 10 co-occurring species for the joint approach
Early detection of range expansion	Presence-only initially, moving to tree based methods as monitoring accumulates	20–50 initially, expanding
Community level impact assessment	Joint species distribution models with environmental and spatial random effects	30+ observations per species across ≥ 40 sites spanning the impact gradient

Note. These pairings are starting points. Method selection should also account for computational resources, available expertise, stakeholder requirements for interpretability, and the spatial and temporal resolution the decision actually needs.

S8 Survey design: coverage, effort and data quality

This appendix gives the extended treatment of the material summarised in Section 3.3 of the main text.

Coverage against total sample size. Spatially dispersed observations spanning the full environmental gradient outperform concentrated samples of equal or larger size (Robinson et al., 2011; Hengl et al., 2018). The mechanism is representation of environmental space: clustered samples leave regions unobserved, so prediction there is extrapolation rather than interpolation, and the model is being asked a question its training data cannot answer regardless of how many records that training data contains.

Stratification. Environmental stratification allocates effort proportionally across the gradients structuring the response, typically sea surface temperature, depth zone and productivity regime. Geographic stratification prioritises breadth across regions and is the practical approximation where the structuring gradients are not known in advance. Temporal stratification samples across seasons and years, capturing phenological variation and supplying the replication that spatio-temporal methods require structurally. Under budget constraint, breadth of coverage should be preferred to intensity of local effort (Canonico et al., 2019).

Effort and detection. Sample quality interacts with quantity. Geostatistical models represent varying effort through the likelihood; tree based methods treat effort as uniform unless it is supplied as a predictor, so an unmodelled effort gradient is absorbed into the fitted habitat relationship. Imperfect detection reduces effective sample size relative to the nominal count, and the reduction is severe where detection probability is low. Taxonomic certainty and covariate matching bear on effective sample size in the same way.

Priorities under constraint. In the order in which they bind: broad spatial coverage ensuring environmental space is represented; sample size meeting the method specific minimum; temporal replication where a spatio-temporal method is intended; and environmental covariates at appropriate resolution and temporal alignment.

Below the threshold. With fewer than roughly 30 observations, practitioners should either accept an exploratory analysis with transparent limitations, or defer formal modelling (Guillera-Arroita et al., 2015). Presence-only methods can provide initial insight framed as hypothesis generating rather than hypothesis testing. Expert derived habitat suitability assessment, built from literature and local knowledge, is a legitimate alternative where data are insufficient. Where modelling proceeds, inference should be restricted to coarse scale pattern with uncertainty stated.

S9 Computational cost

Computational cost bears on method selection wherever prediction must be updated frequently, as in dynamic management (Section 4.2 of the main text). Table S11 gives approximate requirements. Scaling behaviour matters more than absolute timings, since the latter depend on hardware.

Table S11: Approximate computational requirements. Timings assume typical marine datasets of 100 to 10,000 observations with 5 to 15 predictors, and depend on model complexity, spatial resolution and hardware.

Method	Fitting time	Prediction time per surface	Scaling behaviour
GLM	Seconds	Seconds	Linear
GAM ('mgcv')	Seconds to minutes	Seconds	Sub-linear with 'bam'
Random Forest ('ranger')	Seconds to minutes	Seconds	Linear in observations and predictors
BRT ('gbm')	Minutes	Seconds	Linear in tree number
VAST	Minutes to hours	Minutes	Super-linear in knots
sdmTMB	Minutes to ~30 minutes	Seconds to minutes	Super-linear in mesh complexity
tinyVAST	Minutes to hours	Seconds to minutes	Super-linear in mesh complexity and species number
BART	Minutes	Seconds	Linear in tree number
Hmsc	Hours; MCMC with 2–4 chains and 1,000+ posterior samples	Minutes for posterior predictive draws across the prediction grid	Super-linear in species richness and site count
sjSDM	Minutes to ~30 minutes; variational inference	Seconds to minutes	Linear to sub-linear in species count; GPU accelerable

The operational consequence is the one drawn in Section 4.2 of the main text. Tree based methods dominate near real time systems because prediction is fast and requires no mesh, not because they predict better. Where calibrated uncertainty is required in such a system, the cost of a geostatistical alternative falls on fitting rather than prediction, which is why separating frequent field updates from infrequent hyperparameter re-estimation is the plausible route to operational deployment.

S10 Software and implementation

Table S12: Software implementations for the method families in Table 1 of the main text, with the functions that carry the spatial component and notes specific to marine application.

Method	R packages	Key functions and workflow	Python	Notes for marine applications
Random Forest	‘ranger’; ‘randomForest’	<i>ranger()</i> for fast fitting with variable importance; <i>randomForest()</i> as the reference implementation. Spatial predictors are constructed by the analyst	scikit-learn	‘ranger’ preferred above roughly 10,000 observations for its C++ backend. Use permutation importance where predictors are correlated, which is usual with marine covariates
Spatial Random Forest	‘spatialRF’; ‘RFsp’ via ‘landmap’; ‘waywiser’; ‘RandomForestsGLS’	<i>rf_spatial()</i> fits with spatial predictors; <i>rf_evaluate()</i> for spatial evaluation; ‘RandomForestsGLS’ provides the dependent data formulation	verstack	Buffer distances must be computed against an ecologically meaningful reference, not the coordinate grid alone; distance to shore, shelf break or depth contour are the usual choices
VAST	‘VAST’; ‘FishStatsUtils’; ‘TMB’ (Kristensen et al., 2016)	<i>fit_model()</i> is the primary interface; <i>make_data()</i> and <i>make_model()</i> for configuration	limited	Knot number controls both the resolution of the spatial field and the fitting cost; start coarse and refine, since super-linear scaling makes over-specification expensive
sdmTMB	‘sdmTMB’ (Anderson et al., 2025); ‘TMB’; ‘fmesher’	<i>sdmTMB()</i> uses a formula interface comparable to <i>glm</i> or <i>mgcv</i> ; <i>make_mesh()</i> constructs the SPDE mesh	limited	The recommended entry point for practitioners moving from ‘mgcv’. Delta families handle the zero inflation typical of trawl and transect data; <i>extitget_index()</i> produces standardised abundance indices

Method	R packages	Key functions and workflow	Python	Notes for marine applications
tinyVAST	‘tinyVAST’ (Thorson et al., 2025); ‘TMB’; ‘fmesher’	<i>tinyVAST()</i> extends the sdmTMB conventions to multivariate and structural equation specifications	limited	For multi-species work without VAST’s full specification overhead. Cross-species spatial fields borrow strength toward sparsely observed species
Spatial GAMs	‘mgcv’ (Wood, 2017); ‘dsm’ (Miller et al., 2022); ‘gratia’	<i>gam()</i> and <i>bam()</i> with $s(x, y, bs = "tp")$ or <i>te(x, y)</i> ; ‘dsm’ for density surface models	pyGAM	‘dsm’ is the standard route for line transect survey data, which covers most cetacean and seabird applications
MaxEnt	‘maxnet’; ‘ENMeval’; ‘dismo’	<i>maxnet()</i> is the native R implementation requiring no Java; <i>ENMevaluate()</i> automates tuning and background selection	elapid	Background extent defines the domain against which discrimination is measured and must match the area over which prediction is required; it drives apparent performance more than any tuning setting
Boosted Regression Trees	‘gbm’; ‘dismo’; ‘xgboost’; ‘lightgbm’; ‘caret’	<i>gbm.step()</i> performs cross validated selection of tree number with ecologically appropriate defaults (Elith et al., 2008)	xgboost, lightgbm	<i>extitgbm.step()</i> applies ecologically appropriate defaults and cross-validated tree selection; prefer it to direct fitting for distribution work
BART	‘embarcadero’; ‘dbarts’	<i>bart.step()</i> for variable selection in a distribution modelling context; <i>bart()</i> and <i>bart2()</i> for fitting	bartpy	Posterior uncertainty without MCMC cost at community scale; <i>extitbart.step()</i> handles variable selection
Graphical interface	‘GLOSSA’ (Mestre-Tomás et al., 2025)	Shiny application providing a no code interface to BART based models with spatial blocking	—	Useful where analysis must be handed to non-specialist staff, or where a workflow is demonstrated to stakeholders

Method	R packages	Key functions and workflow	Python	Notes for marine applications
Joint SDM, hierarchical	‘Hmsc’	<i>Hmsc()</i> with spatial random effects; <i>createStudyDesign()</i> takes spatial coordinates; <i>Associations()</i> returns residual species correlations. MCMC sampling, hours for large communities	limited	Benefit concentrates where a rare target species co-occurs with better sampled congeners; see Table eftar:jsdm
Joint SDM, variational	‘sjSDM’ (Pichler and Hartig, 2021)	<i>sjSDM()</i> performs fast variational inference through a PyTorch backend	via the same backend	Faster than full MCMC for large communities, at some cost in posterior accuracy; see Table eftar:jsdm
Joint SDM, ordination	‘boral’	<i>boral()</i> for Bayesian latent variable ordination of multivariate abundance data	limited	Ordination and community description rather than spatial prediction; see Table eftar:jsdm
Ensembles	‘biomod2’; ‘caretEnsemble’; ‘stacks’	<i>BIOMOD_Modeling()</i> and <i>BIOMOD_EnsembleModeling()</i> for multi-model platforms	scikit-learn	Appropriate where candidates are inseparable under matched validation (Section 4.1 of the main text), not as a default
Spatial cross validation	‘blockCV’ (Valavi et al., 2019); ‘spatialsample’; ‘ENMeval’	<i>cv_spatial()</i> performs block partitioning with automated range estimation; block size should exceed the residual autocorrelation range	verstack, spav	Block size must exceed the residual autocorrelation range, which should be estimated rather than assumed; <i>extitcv_nndm()</i> suits presence-only data
Spatial diagnostics	‘spdep’; ‘gstat’; ‘DHARMA’	<i>moran.test()</i> and <i>moran.mc()</i> for Moran’s I on residuals; <i>variogram()</i> for empirical variograms; residual simulation for distributional checks	pysal	Moran’s I on residuals rather than on the raw response; residual structure is what bears on the model
Graph neural networks	‘torch’	Limited R support; most implementations use Python	PyTorch Geometric	Noted for completeness; data requirements exceed the settings this review addresses

Table S13: Joint species distribution model frameworks, distinguished by where each applies shrinkage and how inference is performed. The distinction matters under data limitation, because shrinkage on environmental responses is the mechanism by which sparsely observed species borrow strength from better sampled ones.

Framework	R package	Shrinkage on environmental responses	Shrinkage on spatial latent variables	Inference	Recommended marine use
Hierarchical, with spatial random effects	‘Hmsc’	Yes; improves estimation for species with sparse observations	Yes; captures residual spatial autocorrelation across the community	Full MCMC; hours	Demersal fish and benthic assemblages; community level conservation planning where a full posterior is required. Norberg et al. (2019) report the spatial variant outperforming non-spatial joint models
Hierarchical, without spatial random effects	‘Hmsc’	Yes	No	Full MCMC	Only where residual spatial autocorrelation is genuinely weak, or where computation precludes spatial latent factors. Otherwise prefer the spatial variant
Variational	‘sjSDM’ (Pichler and Hartig, 2021)	Yes, through regularisation	Yes, through spatial latent variables	Variational inference; minutes	Large communities above roughly 30 species; iterative assessment against tight timelines. Faster than full MCMC at some cost in posterior accuracy
Bayesian ordination	‘boral’	Yes, through latent variables	No spatial structure in the base implementation	Full MCMC	Compositional analysis and ordination; community description rather than spatial prediction
Multi-species geostatistical	‘VAST’ (Thorson, 2019b)	Implicit, through shared random effects	Yes; vector autoregressive structure on spatial fields across species	Empirical Bayes by Laplace approximation; minutes to hours	Multi-species survey data from structured programmes, and where output feeds an assessment model (Appendix S7)

Framework	R package	Shrinkage on environmental responses	Shrinkage on spatial latent variables	Inference	Recommended marine use
Multi-species geo-statistical, formula interface	‘tinyVAST’ (Thorson et al., 2025)	Implicit, through shared random effects	Yes; cross-species spatial fields with optional simultaneous effects	Empirical Bayes; minutes to hours	For practitioners moving from ‘sdmTMB’ to multi-species work without the full specification overhead

S11 Reporting protocol

We recommend that primary studies contributing to future syntheses report, at minimum: the inference target the validation was designed to estimate; the partitioning scheme and, for spatial blocking, the block size and its relation to the estimated autocorrelation range; at least one discrimination metric and one error or calibration metric where the response type permits both; the spatial and environmental coverage of training data relative to the prediction domain; and the proportion of the prediction domain requiring extrapolation. The ODMAP protocol (Zurell et al., 2020) provides the structure for the first four; the metric convention is the missing element and is the lead priority identified in Section 4.4 of the main text.

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